FICTIONAL TEST PACKET. Not a real application.

Applicant's Name: Avery Chen

STATEMENT OF PURPOSE

I am applying to the Master of Science in Artificial Intelligence at Northwestern because I want to train retrieval systems that clinicians can actually query. For two years I was a research assistant in a clinical natural language processing group. I built the indexing pipeline for a set of de-identified discharge summaries, and I was first author on a workshop paper about that indexer.

The paper, "Sparse Plus Dense Retrieval for Discharge Summaries," appeared in the proceedings of the Upper Midwest NLP Workshop in 2025. I designed the experiment, wrote the retrieval code, and ran the error analysis. The useful result was not a leaderboard number. Dense retrieval missed negation. A hybrid index recovered those cases, and I wrote that up with the counts.

At Northwestern I want the systems and language courses in MSAI, then a project on retrieval for notes where the label is a clinical concept rather than a web query. I have the linear algebra, probability, algorithms, and machine learning sequence from Michigan, and I have already maintained the code other students in the lab ran. I am not looking for a general tour of artificial intelligence. I want a year in which the coursework and a faculty project are the same problem.

EDUCATION

Undergraduate institution: University of Michigan
Degree: B.S. Computer Science, 2025
Cumulative GPA: 3.86

RESEARCH

Research assistant, University of Michigan, September 2023 to May 2025, about 15 hours a week during term and full time in summer 2024. Built a hybrid retrieval index over 80,000 de-identified discharge summaries. First-author workshop paper, Upper Midwest NLP Workshop, 2025. I wrote the code, the experiments, and the first draft.

PROJECTS

Discharge-summary retriever. Python, FAISS, and a small BM25 index. Other students in the lab used the repository for their own error analysis. I reviewed their pull requests.

WORK EXPERIENCE

Undergraduate research assistant, clinical NLP group, University of Michigan, 2023-2025. Teaching assistant for EECS 445, Introduction to Machine Learning, winter 2025, two discussion sections.

HONORS AND AWARDS

EECS 445 teaching award, departmental nominee, 2025. University Honors, 2023-2025.

LETTERS OF RECOMMENDATION

Letter 1. Professor Helen Park, University of Michigan. "Avery is the student I trust with the lab repository. The workshop paper is Avery's experiment, not a footnote on mine. I would fund this student in a PhD program. The retrieval work is careful, and the error analysis is the part I send to new students."

Letter 2. Dr. Omar Shah, University of Michigan. "I co-taught the machine learning course Avery assisted. Avery could explain a confusion matrix to a student who was lost, and then go back and fix a leaking index in the lab. I recommend Avery without reservation for a technical master's program."